Introduction to Data Science and Engineering

- Importing data

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Slide deck originally created by RB Luo

In this lecture

- External data files and file formats
- Work with file system
- Import data files

In this lecture

- **External data files and file formats**
- Work with file system
- Import data files

Why importing data?

- So far, we have worked on built-in datasets. But real-life datasets are everything, everywhere, and come in all kinds of file formats
- Common data file formats
  - `csv`, Comma-Separated Values;
  - `tsv`, Tab-Separated Values;
  - `xls/xlsx`, Microsoft Excel worksheets;
  - `txt`, `json`, `html`: other text-based files;
  - `parquet`, `feather`, `hdf5`, `zarr`: binary data files.
- These formats can be imported to R as data frame easily

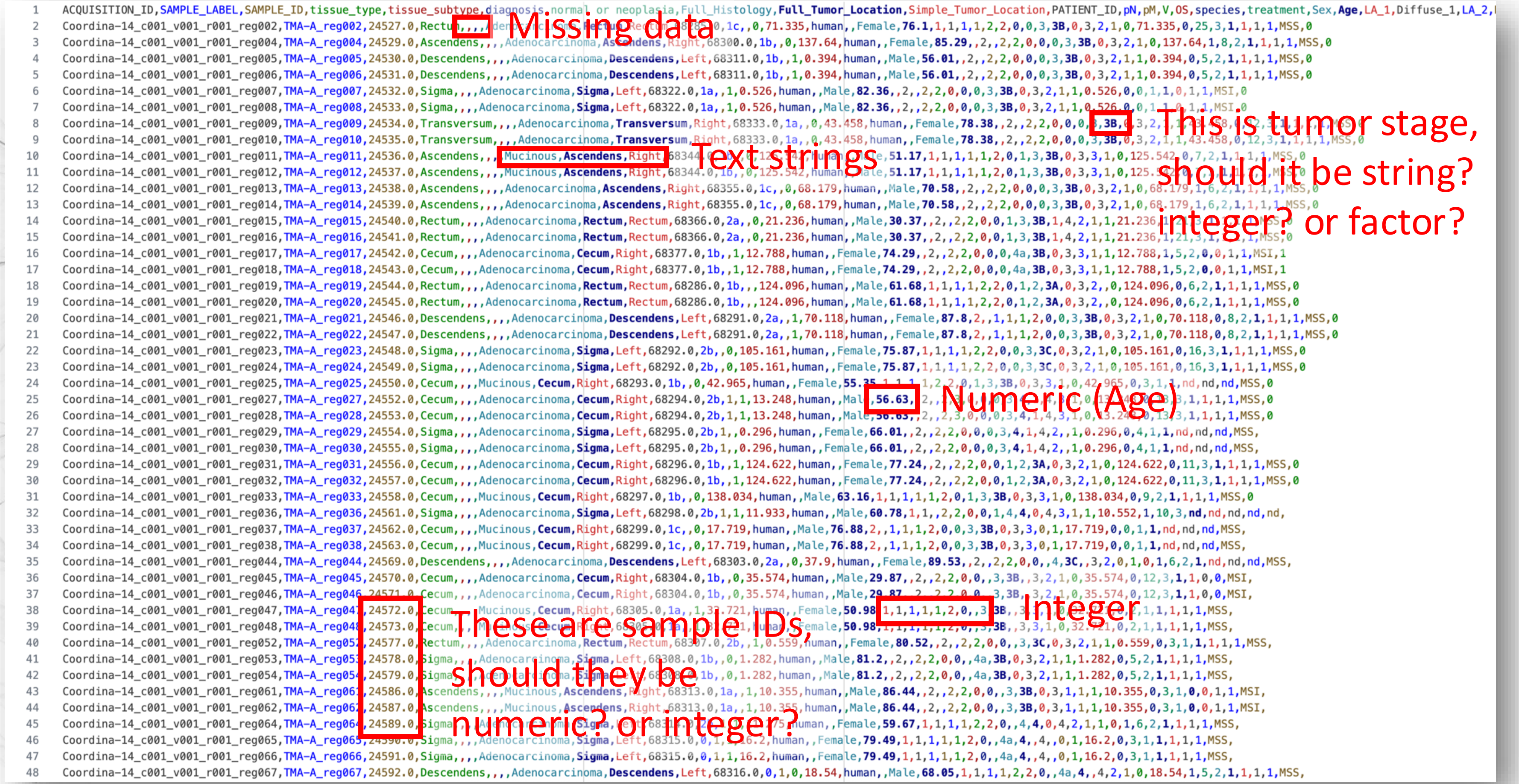
An example csv file

```
ACQUISITION_ID,SAMPLE_LABEL,SAMPLE_ID,tissue_type,tissue_subtype,diagnosis,normal or neoplasia,Full_Histology,Full_Tumor_Location,Simple_Tumor_Location,PATIENT_ID,pN,pM,V,OS,species,treatment,Sex,Age,LA_1,Diffuse_1,LA_2,
Coordina-14_c001_v001_r001_reg002,TMA-A_reg002,24527.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68285.0,1c,,0,71.335,human,,Female,76.1,1,1,1,1,2,2,0,0,3,3B,0,3,2,1,0,71.335,0,25,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg004,TMA-A_reg004,24529.0,Ascendens,,,,Adenocarcinoma,Ascendens,Right,68300.0,1b,,0,137.64,human,,Female,85.29,,2,,2,2,0,0,0,3,3B,0,3,2,1,0,137.64,1,8,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg005,TMA-A_reg005,24530.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1,0.394,human,,Male,56.01,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,0.394,0,5,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg006,TMA-A_reg006,24531.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1,0.394,human,,Male,56.01,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,0.394,0,5,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg007,TMA-A_reg007,24532.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1,0.526,human,,Male,82.36,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,0.526,0,0,1,1,0,1,1,MSI,0
Coordina-14_c001_v001_r001_reg008,TMA-A_reg008,24533.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1,0.526,human,,Male,82.36,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,0.526,0,0,1,1,0,1,1,MSI,0
Coordina-14_c001_v001_r001_reg009,TMA-A_reg009,24534.0,Transversum,,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0,43.458,human,,Female,78.38,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,43.458,0,12,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg010,TMA-A_reg010,24535.0,Transversum,,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0,43.458,human,,Female,78.38,,2,,2,2,0,0,0,3,3B,0,3,2,1,1,43.458,0,12,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg011,TMA-A_reg011,24536.0,Ascendens,,,,Mucinous,Ascendens,Right,68344.0,1b,,0,125.542,human,,Male,51.17,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg012,TMA-A_reg012,24537.0,Ascendens,,,,Mucinous,Ascendens,Right,68344.0,1b,,0,125.542,human,,Male,51.17,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg013,TMA-A_reg013,24538.0,Ascendens,,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0,68.179,human,,Male,70.58,,2,,2,2,0,0,0,3,3B,0,3,2,1,0,68.179,1,6,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg014,TMA-A_reg014,24539.0,Ascendens,,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0,68.179,human,,Male,70.58,,2,,2,2,0,0,0,3,3B,0,3,2,1,0,68.179,1,6,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg015,TMA-A_reg015,24540.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0,21.236,human,,Male,30.37,,2,,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg016,TMA-A_reg016,24541.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0,21.236,human,,Male,30.37,,2,,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg017,TMA-A_reg017,24542.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1,12.788,human,,Female,74.29,,2,,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
Coordina-14_c001_v001_r001_reg018,TMA-A_reg018,24543.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1,12.788,human,,Female,74.29,,2,,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
Coordina-14_c001_v001_r001_reg019,TMA-A_reg019,24544.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,,124.096,human,,Male,61.68,1,1,1,1,2,2,0,1,2,3A,0,3,2,,0,124.096,0,6,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg020,TMA-A_reg020,24545.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,,124.096,human,,Male,61.68,1,1,1,1,2,2,0,1,2,3A,0,3,2,,0,124.096,0,6,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg021,TMA-A_reg021,24546.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1,70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg022,TMA-A_reg022,24547.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1,70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg023,TMA-A_reg023,24548.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0,105.161,human,,Female,75.87,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg024,TMA-A_reg024,24549.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0,105.161,human,,Female,75.87,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg025,TMA-A_reg025,24550.0,Cecum,,,,Mucinous,Cecum,Right,68293.0,1b,,0,42.965,human,,Female,55.35,1,1,1,1,2,2,0,1,3,3B,0,3,3,1,0,42.965,0,3,1,1,nd,nd,nd,MSS,0
Coordina-14_c001_v001_r001_reg027,TMA-A_reg027,24552.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,,2,3,0,0,0,3,4,1,4,3,1,0,13.248,0,13,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg028,TMA-A_reg028,24553.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,,2,3,0,0,0,3,4,1,4,3,1,0,13.248,0,13,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg029,TMA-A_reg029,24554.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,,0.296,human,,Female,66.01,,2,,2,2,0,0,0,3,4,1,4,2,,1,0.296,0,4,1,1,nd,nd,nd,MSS,
Coordina-14_c001_v001_r001_reg030,TMA-A_reg030,24555.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,,0.296,human,,Female,66.01,,2,,2,2,0,0,0,3,4,1,4,2,,1,0.296,0,4,1,1,nd,nd,nd,MSS,
Coordina-14_c001_v001_r001_reg031,TMA-A_reg031,24556.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1,124.622,human,,Female,77.24,,2,,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg032,TMA-A_reg032,24557.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1,124.622,human,,Female,77.24,,2,,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg033,TMA-A_reg033,24558.0,Cecum,,,,Mucinous,Cecum,Right,68297.0,1b,,0,138.034,human,,Male,63.16,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,138.034,0,9,2,1,1,1,1,MSS,0
Coordina-14_c001_v001_r001_reg036,TMA-A_reg036,24561.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68298.0,2b,1,1,11.933,human,,Male,60.78,1,1,,2,2,0,0,1,4,4,0,4,3,1,1,10.552,1,10,3,nd,nd,nd,nd,nd,
Coordina-14_c001_v001_r001_reg037,TMA-A_reg037,24562.0,Cecum,,,,Mucinous,Cecum,Right,68299.0,1c,,0,17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,nd,MSS,
Coordina-14_c001_v001_r001_reg038,TMA-A_reg038,24563.0,Cecum,,,,Mucinous,Cecum,Right,68299.0,1c,,0,17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,nd,MSS,
Coordina-14_c001_v001_r001_reg044,TMA-A_reg044,24569.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68303.0,2a,,0,37.9,human,,Female,89.53,,2,,2,2,0,0,,4,3C,,3,2,0,1,0,1,6,2,1,nd,nd,nd,MSS,
Coordina-14_c001_v001_r001_reg045,TMA-A_reg045,24570.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0,35.574,human,,Male,29.87,,2,,2,2,0,0,,3,3B,,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,
Coordina-14_c001_v001_r001_reg046,TMA-A_reg046,24571.0,Cecum,,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0,35.574,human,,Male,29.87,,2,,2,2,0,0,,3,3B,,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,
Coordina-14_c001_v001_r001_reg047,TMA-A_reg047,24572.0,Cecum,,,,Mucinous,Cecum,Right,68305.0,1a,,1,32.721,human,,Female,50.98,1,1,1,1,1,2,0,,3,3B,,3,3,1,0,32.721,0,2,1,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg048,TMA-A_reg048,24573.0,Cecum,,,,Mucinous,Cecum,Right,68305.0,1a,,1,32.721,human,,Female,50.98,1,1,1,1,1,2,0,,3,3B,,3,3,1,0,32.721,0,2,1,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg052,TMA-A_reg052,24577.0,Rectum,,,,Adenocarcinoma,Rectum,Rectum,68307.0,2b,,1,0.559,human,,Female,80.52,,2,,2,2,0,0,,3,3C,0,3,2,1,1,0.559,0,3,1,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg053,TMA-A_reg053,24578.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0,1.282,human,,Male,81.2,,2,,2,2,0,0,,4a,3B,0,3,2,1,1,1.282,0,5,2,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg054,TMA-A_reg054,24579.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0,1.282,human,,Male,81.2,,2,,2,2,0,0,,4a,3B,0,3,2,1,1,1.282,0,5,2,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg061,TMA-A_reg061,24586.0,Ascendens,,,,Mucinous,Ascendens,Right,68313.0,1a,,1,10.355,human,,Male,86.44,,2,,2,2,0,0,,3,3B,0,3,1,1,1,10.355,0,3,1,0,0,1,1,MSI,
Coordina-14_c001_v001_r001_reg062,TMA-A_reg062,24587.0,Ascendens,,,,Mucinous,Ascendens,Right,68313.0,1a,,1,10.355,human,,Male,86.44,,2,,2,2,0,0,,3,3B,0,3,1,1,1,10.355,0,3,1,0,0,1,1,MSI,
Coordina-14_c001_v001_r001_reg064,TMA-A_reg064,24589.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68314.0,2b,1,1,11.275,human,,Female,59.67,1,1,1,1,2,2,0,,4,4,0,4,2,1,1,0,1,6,2,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg065,TMA-A_reg065,24590.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68315.0,0,1,1,16.2,human,,Female,79.49,1,1,1,1,1,2,0,,4a,4,,4,,0,1,16.2,0,3,1,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg066,TMA-A_reg066,24591.0,Sigma,,,,Adenocarcinoma,Sigma,Left,68315.0,0,1,1,16.2,human,,Female,79.49,1,1,1,1,1,2,0,,4a,4,,4,,0,1,16.2,0,3,1,1,1,1,1,MSS,
Coordina-14_c001_v001_r001_reg067,TMA-A_reg067,24592.0,Descendens,,,,Adenocarcinoma,Descendens,Left,68316.0,0,1,0,18.54,human,,Male,68.05,1,1,1,1,2,2,0,,4a,4,,4,2,1,0,18.54,1,5,2,1,1,1,1,MSS,
```

A dataset of colorectal cancer patients, each row represents a patient, with:

- Tumor type, location, grading and stage, demographics, etc.

An example csv file



A dataset of colorectal cancer patients, each row represents a patient, with:

- Tumor type, location, grading and stage, demographics, etc.

`murders` dataset in a CSV file

The table loaded by `data(murders)` can be downloaded from
https://raw.githubusercontent.com/rafalab/dslabs/master/inst/extdata/murders.csv

```
> library(dslabs)
> data(murders)
```

```
state,abb,region,population,total
Alabama,AL,South,4779736,135
Alaska,AK,West,710231,19
Arizona,AZ,West,6392017,232
Arkansas,AR,South,2915918,93
California,CA,West,37253956,1257
Colorado,CO,West,5029196,65
Connecticut,CT,Northeast,3574097,97
Delaware,DE,South,897934,38
District of Columbia,DC,South,601723,99
```

The first row (i.e., header) contains column names. Most data import functions assume presence of a header. How to check?

- In R: `read_lines("murders.csv", n_max = 3)`
- In shell: `head murders.csv -n 3`

Following rows contain actual table contents, note that ‘`,`’ serve as delimiter between columns.

- Delimiters can also be other symbols such as semicolon ‘`;`’
- You can specify which delimiter to look for when parsing files.

In this lecture

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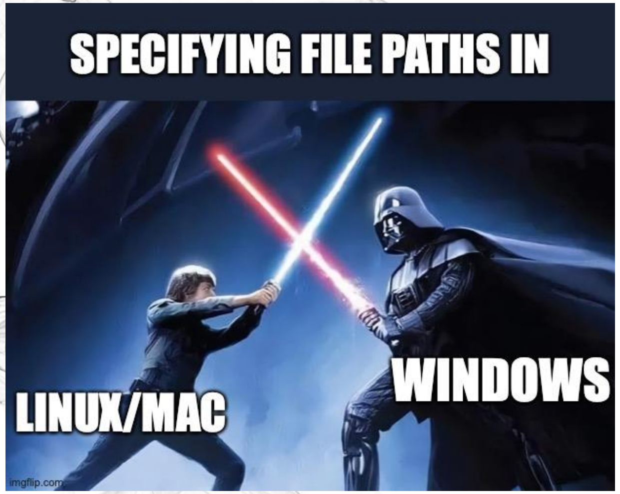
- **Work with file system**
- Import data files

Relative and Absolute Paths



- Absolute path: the complete location of a file/folder from the root.
  - Linux/Mac: starting from '`/`'
  - Windows: starting from '`C:\`', '`D:\`', etc., note that forward slash (`/`) usually works for Windows paths (in Rstudio) as well;

```
> system.file(package = "dslabs")
[1] "/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/dslabs"
```

- Relative path: anything that doesn't start from the root
  - `file_in_this_folder.txt`
  - `subdir/file_in_subdir.txt`
  - `../file_in_parent_folder.txt`
- In R:
  - `file.path(DIR1, DIR2, FILE_NAME)`: for composing paths, no more `/\` confusions;
  - `normalizePath(RELATIVE_PATH)`: converts relative path to absolute path;
  - `list.files(DIR)`: lists files in a folder;
  - `file.exists(PATH)`: checks if a file exists at the specified path.

Working Directory

- Relative path is generally more recommended, as absolute paths may vary across devices, platforms, etc., making them less portable
- Preferred approach: organize all your data files, codes, and markdown/notebook files in the same place (i.e., working directory)

```
> getwd()
[1] "/Users/zqwu"
> dir.create("CS2501_demo")
> setwd("/Users/zqwu/CS2501_demo/")
> filename <- "murders.csv"
> dir <- system.file("extdata", package = "dslabs")
> data_file_path <- file.path(dir, filename)
> file.copy(data_file_path, "murders.csv")
[1] TRUE
> list.files()
[1] "murders.csv"
```

- Use `getwd()`, `setwd(PATH)` to view/change your working directory
- Move your data files to your working directory
- Share the entire working directory when communicating results, data, codes, etc.

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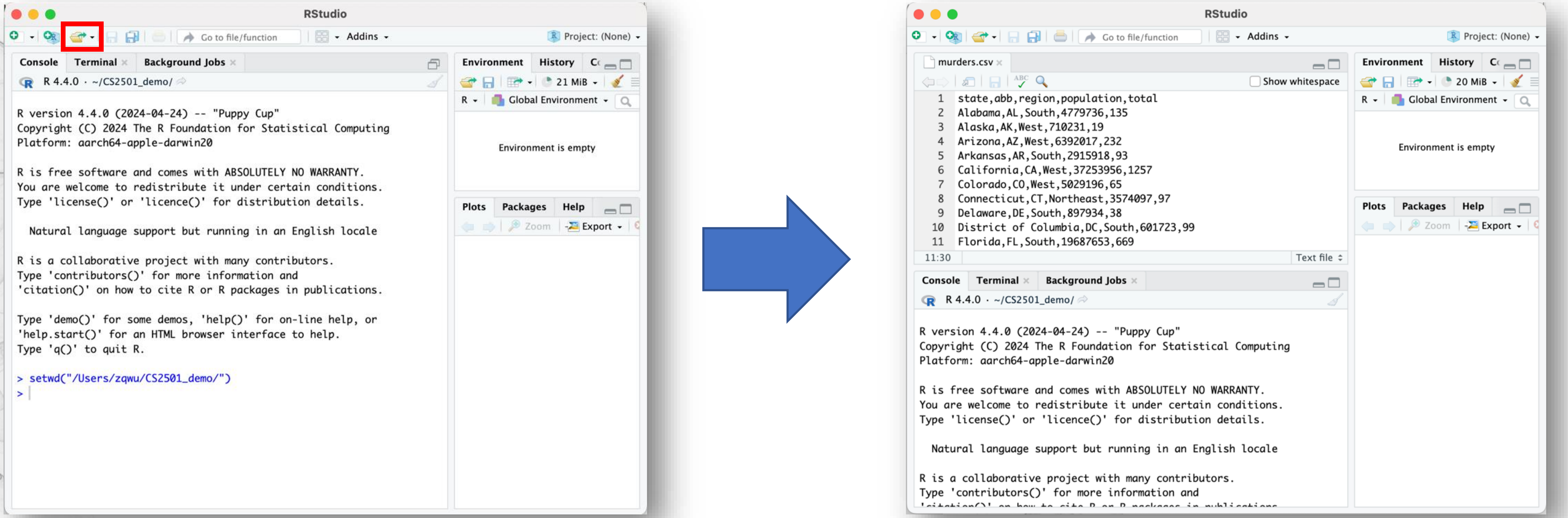
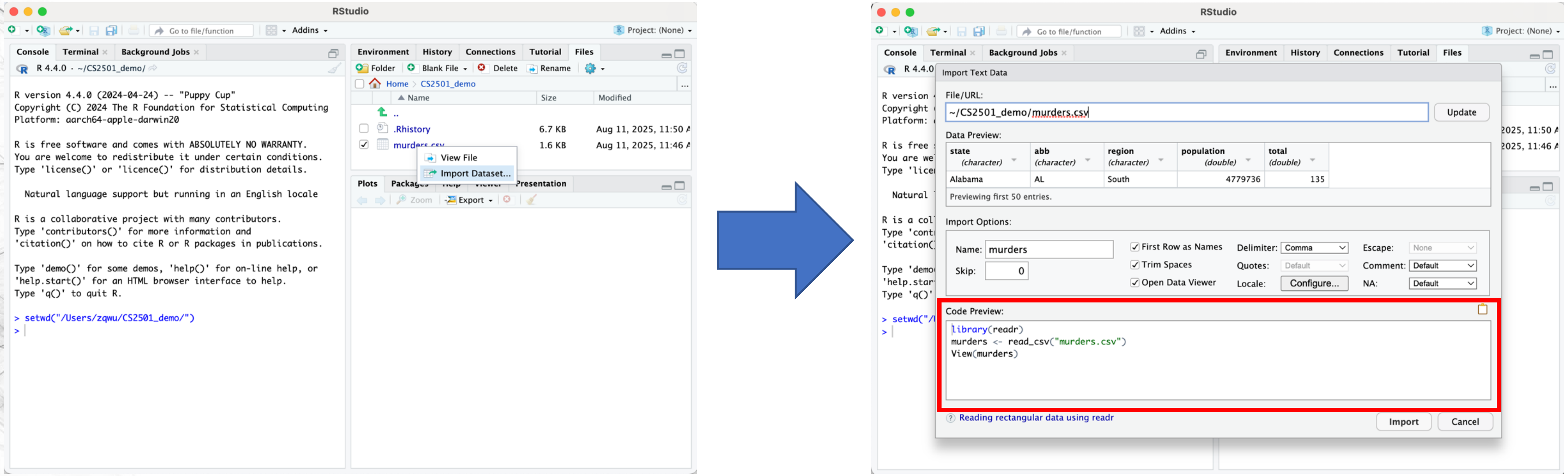
- **Import data files**

Read data file using Rstudio GUI



- This is opening the data file in your editor, not importing it to your environment

Read data file using Rstudio GUI



- Not recommended, use codes when you can

read_csv

```
> library(readr)
> murders <- read_csv("murders.csv")
Rows: 51 Columns: 5
── Column specification ──────────────────
Delimiter: ","
chr (3): state, abb, region
dbl (2): population, total

ℹ Use `spec()` to retrieve the full column
s data.
ℹ Specify the column types or set `show_col
iet this message.
> head(murders, 3)
# A tibble: 3 × 5
  state   abb   region population total
  <chr>   <chr> <chr>       <dbl> <dbl>
1 Alabama AL    South     4779736   135
2 Alaska  AK    West       710231    19
3 Arizona AZ    West      6392017   232
```

```
> spec(murders)
cols(
  state = col_character(),
  abb = col_character(),
  region = col_character(),
  population = col_double(),
  total = col_double()
)
```

- Use `read_csv` to read a local file
- See also `?read_csv` for its optional arguments

read_csv

```
> covid <- read_csv("http://www.bio8.cs.hku.hk/comp2501/covid.csv")
Rows: 47480 Columns: 12
── Column specification ────────────────────────────────────────────
Delimiter: ","
chr (5): dateRep, countriesAndTerritories, geoId, countryterritoryCode, contine...
dbl (7): day, month, year, cases, deaths, popData2019, Cumulative_number_for_14...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
> head(covid)
# A tibble: 6 × 12
  dateRep      day month  year cases deaths countriesAndTerritories geoId
  <chr>      <dbl> <dbl> <dbl> <dbl>  <dbl> <chr>                   <chr>
1 07/10/2020     7    10  2020    62      2 Afghanistan             AF
2 06/10/2020     6    10  2020   145      5 Afghanistan             AF
3 05/10/2020     5    10  2020    44      0 Afghanistan             AF
4 04/10/2020     4    10  2020     7      4 Afghanistan             AF
5 03/10/2020     3    10  2020     5      0 Afghanistan             AF
6 02/10/2020     2    10  2020    17      0 Afghanistan             AF
# ℹ 4 more variables: countryterritoryCode <chr>, popData2019 <dbl>,
#   continentExp <chr>,
#   `Cumulative_number_for_14_days_of_COVID-19_cases_per_100000` <dbl>
> spec(covid)
cols(
  dateRep = col_character(),
  day = col_double(),
  month = col_double(),
  year = col_double(),
  cases = col_double(),
  deaths = col_double(),
  countriesAndTerritories = col_character(),
  geoId = col_character(),
  countryterritoryCode = col_character(),
  popData2019 = col_double(),
  continentExp = col_character(),
  `Cumulative_number_for_14_days_of_COVID-19_cases_per_100000` = col_double()
)
```

http://www.bio8.cs.hku.hk/comp2501/covid.csv

- You can also use `read_csv` to (download and) read a csv file from url
- `download.file(url, "covid.csv")` for explicitly downloading a copy

Other file readers

`read_csv` is in the `readr` library (part of `tidyverse`)
There are some other functions that suit different file format

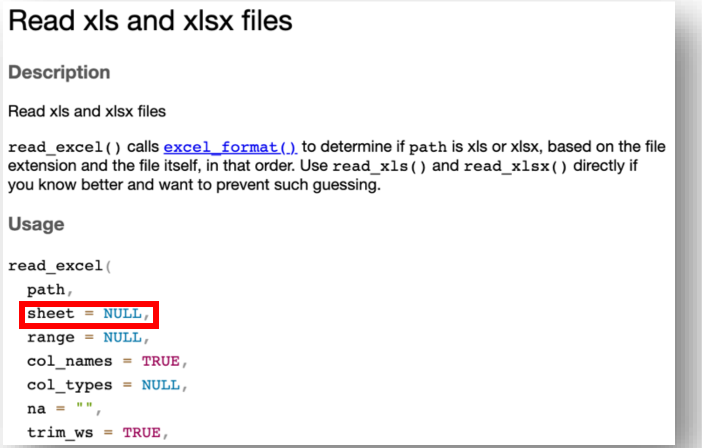
Function	Format	Typical suffix
read_table	white space separated values	txt
read_csv	comma separated values	csv
read_csv2	semicolon separated values	csv
read_tsv	tab delimited separated values	tsv
read_delim	general text file format, must define delimiter	txt

Hey, how to read Microsoft Excel?

`readxl` library

Function	Format	Typical suffix
read_excel	auto detect the format	xls, xlsx
read_xls	original format	xls
read_xlsx	new format	xlsx

The Microsoft Excel formats permit you to have more than one spreadsheet in one file, referred to as sheets.
The functions listed above read the **FIRST** sheet by default.
To read other sheets, use names of the sheets for the `sheet` argument.



Read xls and xlsx files

Description

Read xls and xlsx files

`read_excel()` calls `excel_format()` to determine if `path` is xls or xlsx, based on the file extension and the file itself, in that order. Use `read_xls()` and `read_xlsx()` directly if you know better and want to prevent such guessing.

Usage

```
read_excel(
  path,
  sheet = NULL,
  range = NULL,
  col_names = TRUE,
  col_types = NULL,
  na = "",
  trim_ws = TRUE,
```

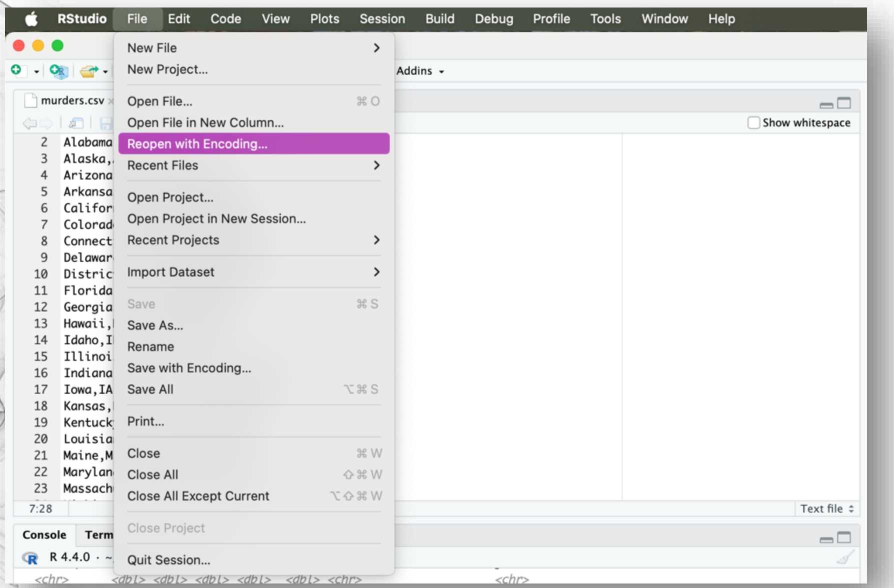
Encoding

- Encoding: text => bit vector

Encoding	Description	Character Set	Size	Use Case
ASCII	Original character encoding standard.	English letters, digits, and basic symbols (128 characters).	1 byte/char	Legacy files, simple text files.
UTF-8	Most widely used Unicode encoding; variable length; backward compatible with ASCII.	All Unicode characters.	1–4 bytes/char	Default for modern systems, web applications, and multilingual text.
UTF-16	Variable-length Unicode encoding; **not** directly compatible with ASCII.	All Unicode characters.	2 or 4 bytes/char	Multilingual text, applications requiring wide character sets, e.g., Windows APIs.
GBK & GB18030	Simplified and traditional Chinese encoding (GB18030 is a superset of GBK).	~21,000 Chinese characters (GBK); all Unicode characters (GB18030).	2 bytes/char (GBK); 1–4 bytes/char (GB18030).	Simplified Chinese text files (GBK); full Unicode support (GB18030).
Big5 & Big5-HKSCS	Traditional Chinese encoding used in Taiwan, Hong Kong, etc.	~13,000 Traditional Chinese characters (Big5); additional characters in Big5-HKSCS.	2 bytes/char	Traditional Chinese text files.

- RStudio defaults to UTF-8

Encoding




```
> read_csv(file_path)
Rows: 3 Columns: 4
Error in nchar(x, "width") : invalid multibyte string, element 1
> read_csv(file_path, locale = locale(encoding = "GB18030"))
Rows: 3 Columns: 4
── Column specification ──────────────────────────────────────────
Delimiter: ","
chr (3): 姓, 名, 城市
dbl (1): 年龄

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
# A tibble: 3 × 4
  姓    名     年龄 城市
  <chr> <chr> <dbl> <chr>
1 张    三       25 北京
2 李    四       30 上海
3 王    五       28 广州
```

Good habits of using Excel/Spreasheet to process your data

- Be Consistent
- Choose Good Names for Things
- Write Dates as YYYY-MM-DD
- No Empty Cells
- Put Just One Thing in a Cell
- Make It a Rectangle
- Create a Data Dictionary
- No Calculations in the Raw Data Files
- Do Not Use Font Color or Highlighting as Data
- Make Backups
- Use Data Validation to Avoid Errors
- Save the Data as Text Files

See also: https://rafalab.dfci.harvard.edu/dsbook/importing-data.html#organizing-data-with-spreadsheets

In this lecture

- External data files and file formats
- Work with file system
  - Relative path vs absolute path
  - Working directory
- Import data files
  - `readr`: `read_csv`, etc.
  - `readxl`: `read_excel`, etc.
  - Encodings